

```

import numpy as np
import matplotlib.pyplot as plt

# Datos
x = np.array([0, 0, 2, 2])
y = np.array([0, 1, 2, 1])

# Minimos Cuadrados
m, b = np.polyfit(x, y, 1)

print("m:", m)
print("b:", b)

# Ecuación
print(f"Ecuación: y = {m:.2f}x + {b:.2f}")

y_pred = m * x + b

plt.scatter(x, y, label="Datos")
plt.plot(x, y_pred, label="Recta de ajuste")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()

# Datos
x_1 = np.array([10, 5, 0])
y_2 = np.array([1, 2, 3])

m_1, b_2 = np.polyfit(x_1, y_2, 1)

print("m:", m_1)
print("b:", b_2)

# Ecuación
print(f"Ecuación: y = {m_1:.2f}x + {b_2:.2f}")

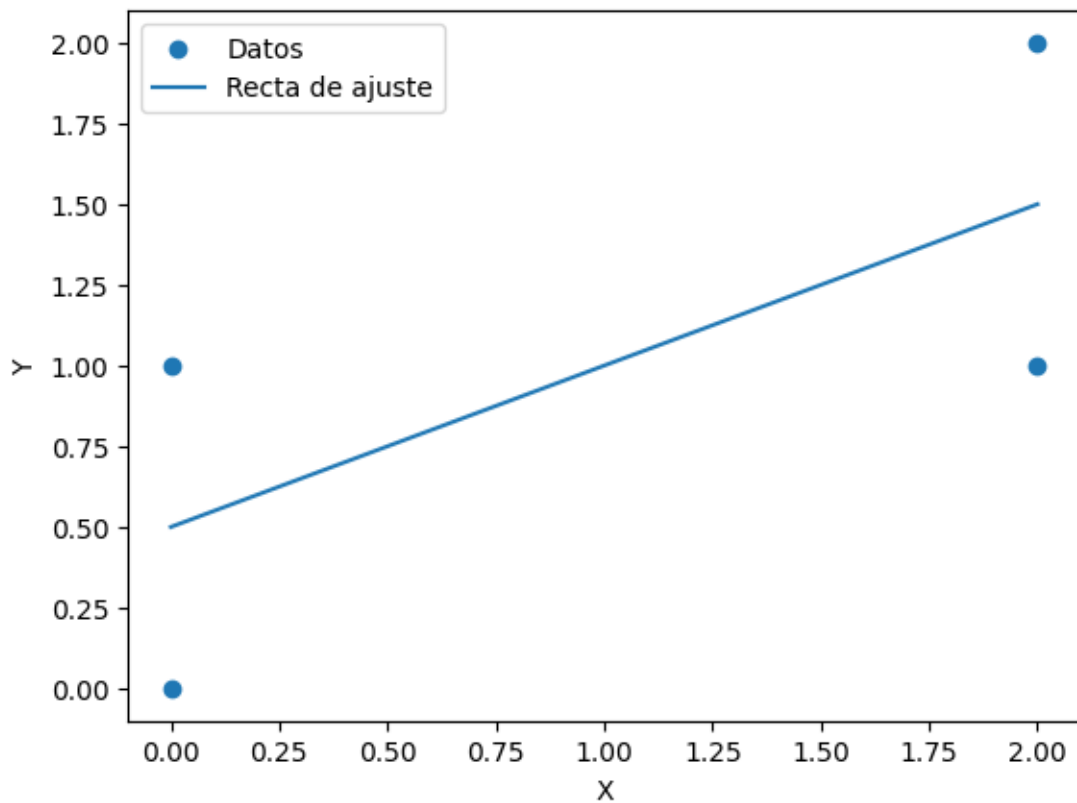
y_pred_1 = m_1 * x_1 + b_2

plt.scatter(x_1, y_2, label="Datos")
plt.plot(x_1, y_pred_1, label="Recta")
plt.xlabel("X")

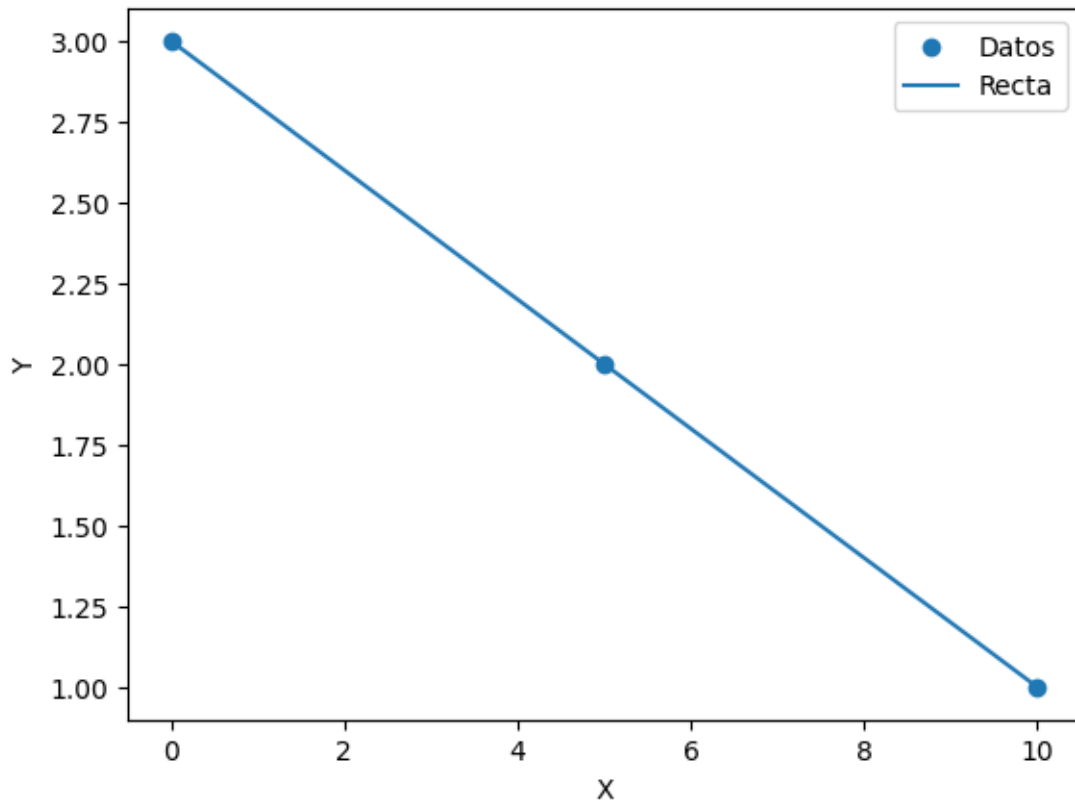
```

```
plt.ylabel("Y")
plt.legend()
plt.show()
```

m: 0.5000000000000002
b: 0.5000000000000001
Ecuación: $y = 0.50x + 0.50$



m: -0.19999999999999996
b: 2.9999999999999999
Ecuación: $y = -0.20x + 3.00$



```
import numpy as np
import matplotlib.pyplot as plt

x_3 = np.array([
    3.38, 0.35, 2.07, -0.45, -0.55,
    -1.06, -2.77, 3.08, -3.98, -5.0,
    -3.18, -1.96, 2.37, -1.66, 4.09
])

y_3 = np.array([
    -0.04, 1.28, 0.65, 2.18, 1.70,
    1.96, 3.36, 0.18, 3.35, 4.16,
    2.95, 2.34, 0.38, 1.75, -1.06
])

a, b, c = np.polyfit(x_3, y_3, 2)

print("a =", a)
```

```

print("b =", b)
print("c =", c)

print(f"Ecuación:  $y = \{a:.4f\}x^2 + \{b:.4f\}x + \{c:.4f\}$ ")

x_line = np.linspace(min(x_3), max(x_3), 100)
y_line = a*x_line**2 + b*x_line + c

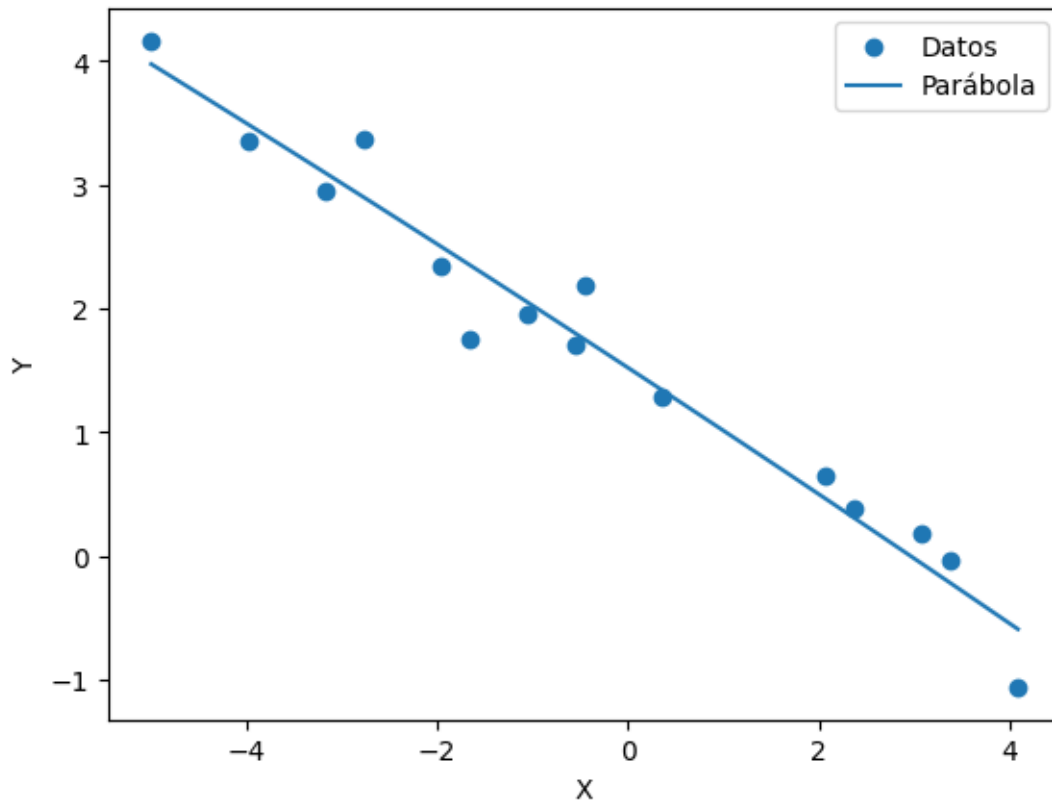
plt.scatter(x_3, y_3, label="Datos")
plt.plot(x_line, y_line, label="Parábola")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()

```

```

a = -0.002845009119942713
b = -0.5046543047259151
c = 1.5204178119974625
Ecuación:  $y = -0.0028x^2 + -0.5047x + 1.5204$ 

```



```
import numpy as np
import matplotlib.pyplot as plt

xs = ([
    3.38, 0.35, 2.07, -0.45, -0.55,
    -1.06, -2.77, 3.08, -3.98, -5.0,
    -3.18, -1.96, 2.37, -1.66, 4.09
])

ys = ([
    -0.04, 1.28, 0.65, 2.18, 1.70,
    1.96, 3.36, 0.18, 3.35, 4.16,
    2.95, 2.34, 0.38, 1.75, -1.06
])

def lineal(xs, ys):

    xs = np.array(xs, dtype=float)
```

```

ys = np.array(ys, dtype=float)

sum_y = np.sum(ys)
sum_x = np.sum(xs)

n = len(xs)

sum_xy = 0
sum_x2 = 0

for x, y in zip(xs, ys):
    sum_xy += x * y
    sum_x2 += x**2

m = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x**2)

b = (sum_y - m * sum_x) / n

y_pred = m * xs + b

errores_abs = np.abs(ys - y_pred)

error_promedio = np.mean(errores_abs)
desviacion = np.std(errores_abs)

idx_max = np.argmax(errores_abs)
idx_min = np.argmin(errores_abs)

print(f"y = {m:.4f}x + {b:.4f}")
print("Error absoluto promedio:", error_promedio)
print("Desviación estándar:", desviacion)

print("\nError máximo")
print(xs[idx_max], ys[idx_max])

print("\nError mínimo")
print(xs[idx_min], ys[idx_min])

plt.figure(figsize=(8,5))

plt.scatter(xs, ys, label="Datos")
plt.plot(xs, y_pred, label="Lineal")

```

```

plt.scatter(xs[idx_max], ys[idx_max],
            s=150,
            marker='x',
            color='red',
            label='Error máximo')

plt.scatter(xs[idx_min], ys[idx_min],
            s=150,
            marker='o',
            color='green',
            label='Error mínimo')

plt.legend()
plt.show()

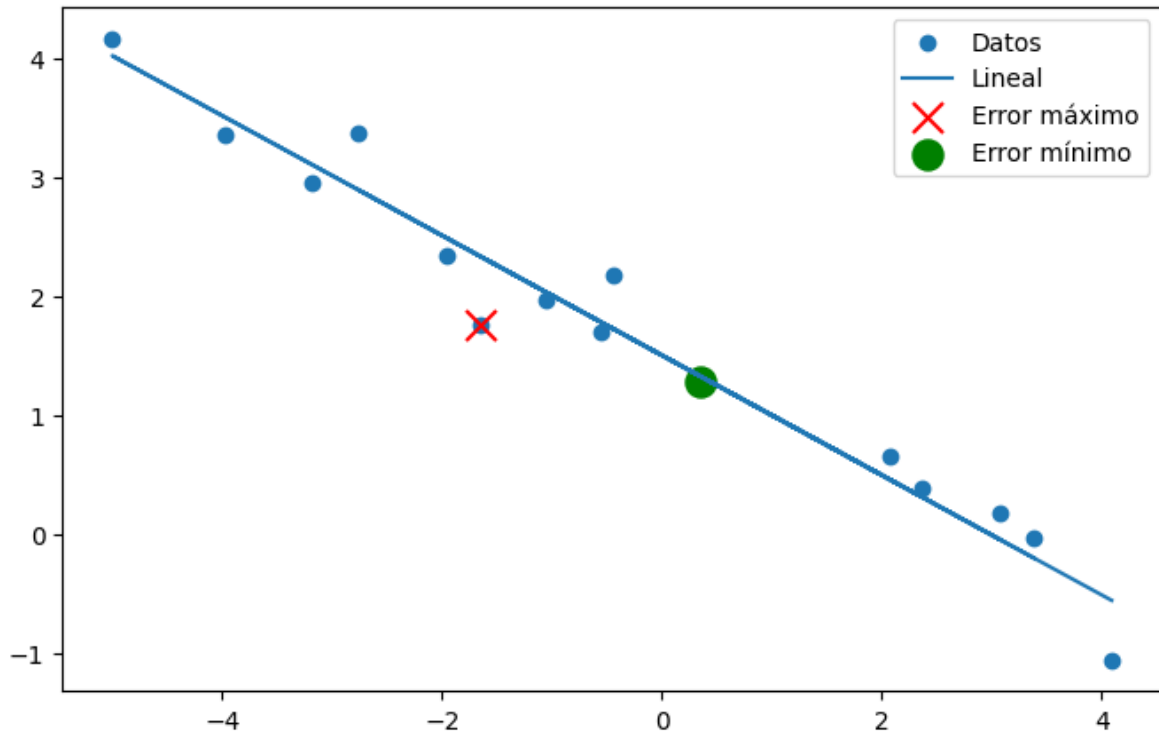
print(lineal(xs, ys))

```

$y = -0.5032x + 1.4992$
 Error absoluto promedio: 0.22990400010035822
 Desviación estándar: 0.17254145556663417

Error máximo
 -1.66 1.75

Error mínimo
 0.35 1.28



None

```
import numpy as np

xs = ([
    3.38, 0.35, 2.07, -0.45, -0.55,
    -1.06, -2.77, 3.08, -3.98, -5.0,
    -3.18, -1.96, 2.37, -1.66, 4.09
])

ys = ([
    -0.04, 1.28, 0.65, 2.18, 1.70,
    1.96, 3.36, 0.18, 3.35, 4.16,
    2.95, 2.34, 0.38, 1.75, -1.06
])

def cuadrada(xs, ys):

    xs = np.array(xs, dtype=float)
    ys = np.array(ys, dtype=float)
```



```

n = len(xs)

sx = np.sum(xs)
sy = np.sum(ys)

sx2 = sx3 = sx4 = 0
sxy = sx2y = 0

for x, y in zip(xs, ys):
    sx2 += x**2
    sx3 += x**3
    sx4 += x**4
    sxy += x*y
    sx2y += (x**2)*y

A = np.array([
    [sx2, sx, n],
    [sx3, sx2, sx],
    [sx4, sx3, sx2]
])

B = np.array([
    sy,
    sxy,
    sx2y
])

a, b, c = np.linalg.solve(A, B)

y_pred = a*xs**2 + b*xs + c

errores_abs = np.abs(ys - y_pred)

error_promedio = np.mean(errores_abs)
desviacion = np.std(errores_abs)

idx_max = np.argmax(errores_abs)
idx_min = np.argmin(errores_abs)

print(f"y = {a:.4f}x2 + {b:.4f}x + {c:.4f}")

print("Error absoluto promedio:", error_promedio)

```

```

print("Desviación estándar:", desviacion)

print("\nError máximo")
print(xs[idx_max], ys[idx_max])

print("\nError mínimo")
print(xs[idx_min], ys[idx_min])

x_line = np.linspace(min(xs), max(xs), 300)
y_line = a*x_line**2 + b*x_line + c

plt.figure(figsize=(8,5))

plt.scatter(xs, ys, label="Datos")
plt.plot(x_line, y_line, label="Cuadrática")

plt.scatter(xs[idx_max], ys[idx_max],
            s=150,
            marker='x',
            color='red',
            label='Error máximo')

plt.scatter(xs[idx_min], ys[idx_min],
            s=150,
            marker='o',
            color='green',
            label='Error mínimo')

plt.legend()
plt.show()

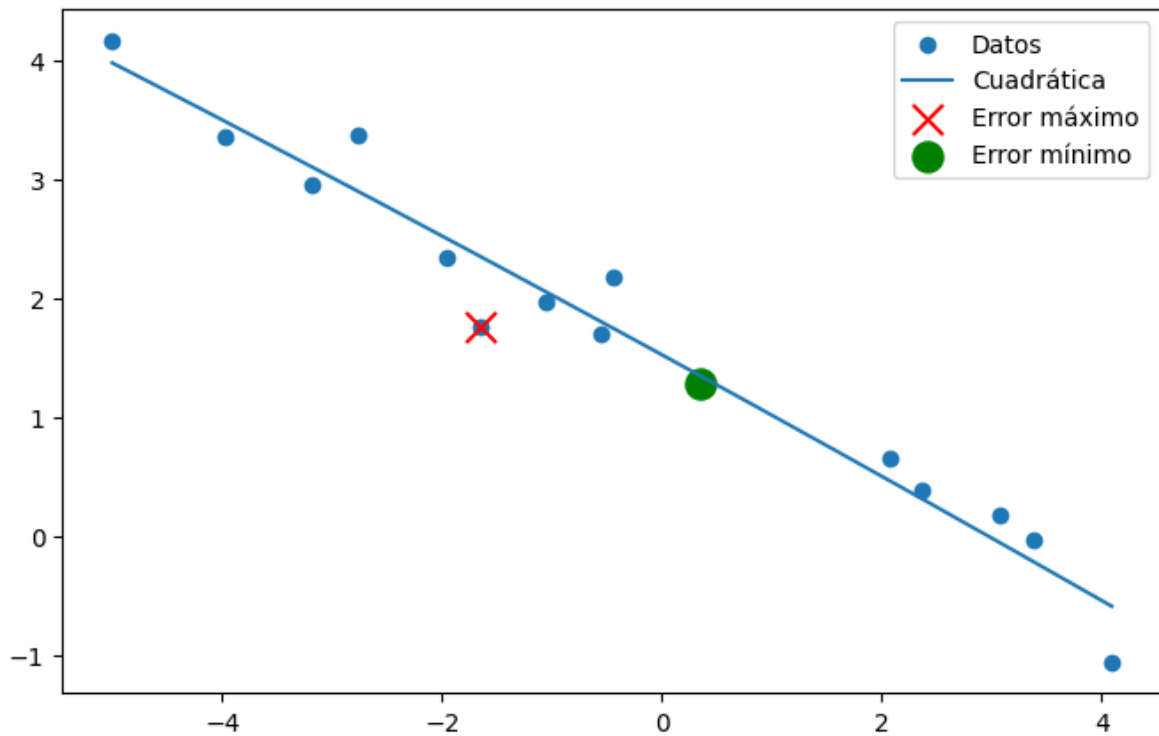
print(cuadrada(xs,ys))

```

$y = -0.0028x^2 + -0.5047x + 1.5204$
 Error absoluto promedio: 0.23476557246855057
 Desviación estándar: 0.16470919437418624

Error máximo
 -1.66 1.75

Error mínimo
 0.35 1.28



None

```
import numpy as np

xs = ([
    3.38, 0.35, 2.07, -0.45, -0.55,
    -1.06, -2.77, 3.08, -3.98, -5.0,
    -3.18, -1.96, 2.37, -1.66, 4.09
])

ys = ([
    -0.04, 1.28, 0.65, 2.18, 1.70,
    1.96, 3.36, 0.18, 3.35, 4.16,
    2.95, 2.34, 0.38, 1.75, -1.06
])

def cubica(xs, ys):

    xs = np.array(xs, dtype=float)
    ys = np.array(ys, dtype=float)
```

```

n = len(xs)

sx = np.sum(xs)
sy = np.sum(ys)

sx2 = sx3 = sx4 = sx5 = sx6 = 0
sxy = sx2y = sx3y = 0

for x, y in zip(xs, ys):

    sx2 += x**2
    sx3 += x**3
    sx4 += x**4
    sx5 += x**5
    sx6 += x**6

    sxy += x*y
    sx2y += (x**2)*y
    sx3y += (x**3)*y

A = np.array([
    [sx3, sx2, sx, n],
    [sx4, sx3, sx2, sx],
    [sx5, sx4, sx3, sx2],
    [sx6, sx5, sx4, sx3]
])

B = np.array([
    sy,
    sxy,
    sx2y,
    sx3y
])

a, b, c, d = np.linalg.solve(A, B)

y_pred = a*xs**3 + b*xs**2 + c*xs + d

errores_abs = np.abs(ys - y_pred)

error_promedio = np.mean(errores_abs)
desviacion = np.std(errores_abs)

```

```

idx_max = np.argmaxerrores_abs)
idx_min = np.argminerrores_abs)

print(f"y = {a:.4f}x3 + {b:.4f}x2 + {c:.4f}x + {d:.4f}")

print("Error absoluto promedio:", error_promedio)
print("Desviación estándar:", desviacion)

print("\nError máximo")
print(xs[idx_max], ys[idx_max])

print("\nError mínimo")
print(xs[idx_min], ys[idx_min])

x_line = np.linspace(min(xs), max(xs), 300)
y_line = a*x_line**3 + b*x_line**2 + c*x_line + d

plt.figure(figsize=(8,5))

plt.scatter(xs, ys, label="Datos")
plt.plot(x_line, y_line, label="Cúbica")

plt.scatter(xs[idx_max], ys[idx_max],
            s=150,
            marker='x',
            color='red',
            label='Error máximo')

plt.scatter(xs[idx_min], ys[idx_min],
            s=150,
            marker='o',
            color='green',
            label='Error mínimo')

plt.legend()
plt.show()

print(cubica(xs,ys))

```

$y = -0.0082x^3 + -0.0162x^2 + -0.3967x + 1.6075$
 Error absoluto promedio: 0.1850019467861469

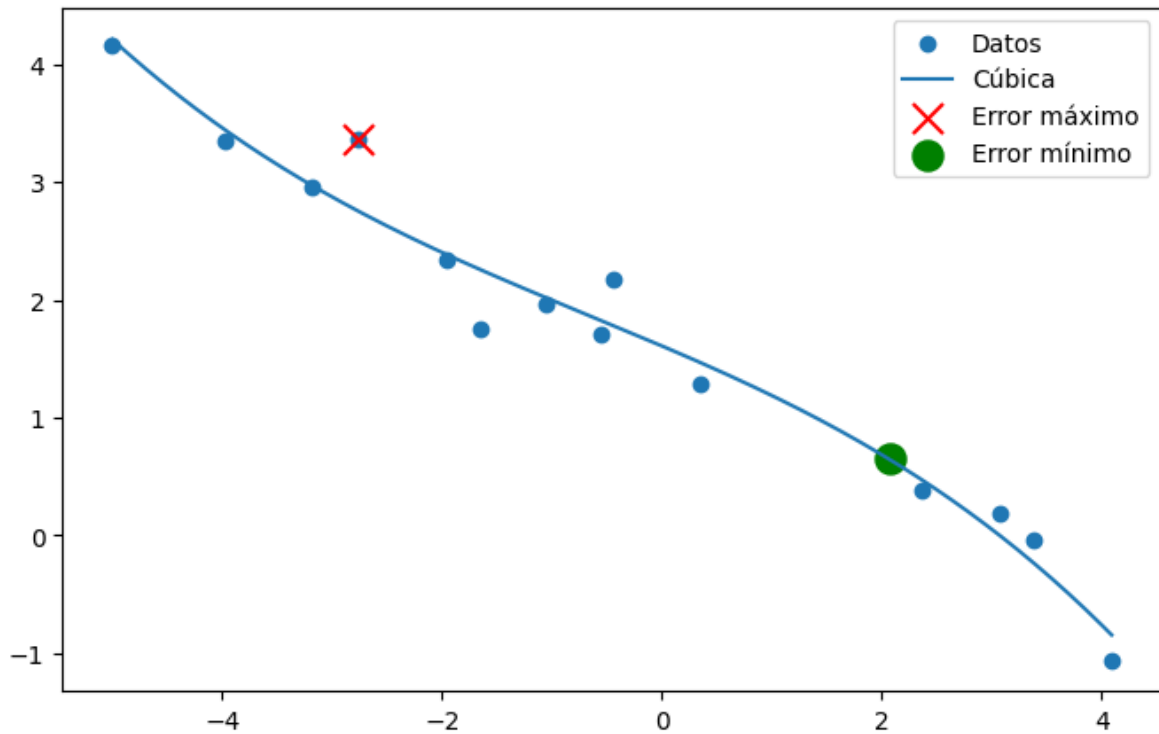
Desviación estándar: 0.17529501044732929

Error máximo

-2.77 3.36

Error mínimo

2.07 0.65



None

```
puntos = [  
    (3.38, 15.65), (0.35, -11.20), (2.07, -4.00), (-0.45, 18.03),  
    (-0.56, 7.94), (-1.06, 15.32), (-2.78, -4.40), (3.08, 1.39),  
    (-3.99, -11.92), (-5.00, -90.24), (-3.18, 6.92), (-1.97, 28.35),  
    (2.37, -2.41), (-1.67, -5.47), (4.09, 35.91), (-4.60, -55.53),  
    (2.68, 0.77), (2.78, 4.79), (-3.79, -27.05), (-1.87, 2.85)  
]  
  
xs = []  
ys = []
```

```

for punto in puntos:
    xs.append(punto[0])
    ys.append(punto[1])

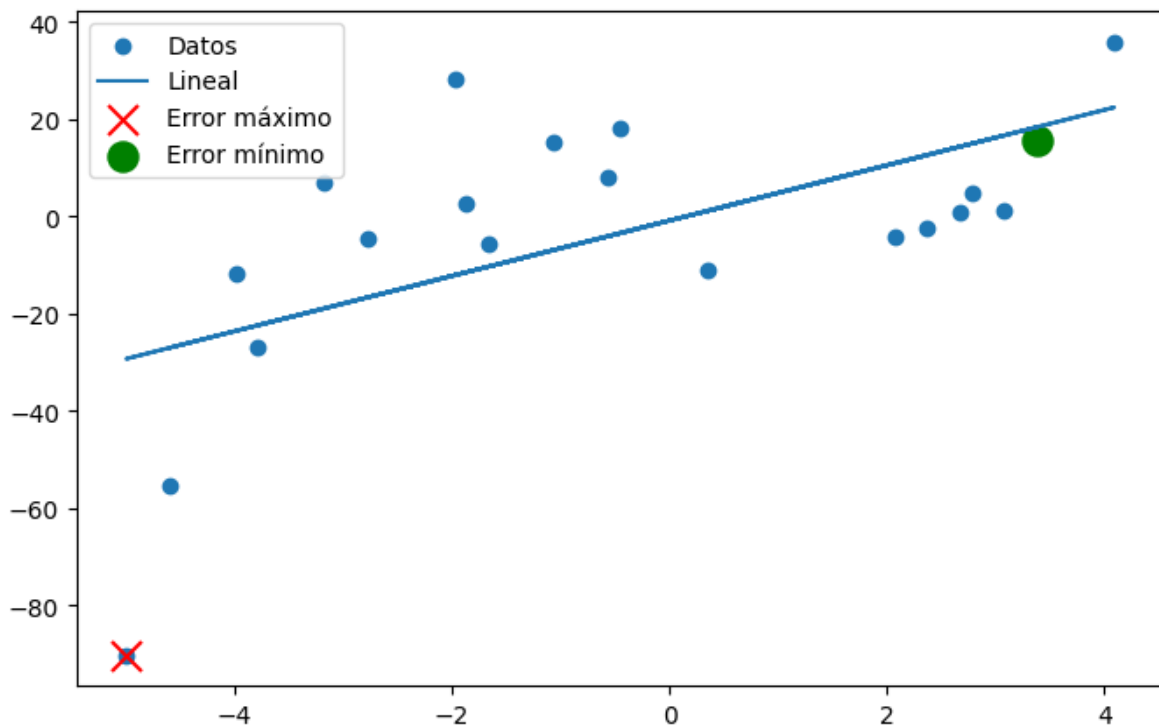
print(lineal(xs, ys))
print(cuadrada(xs, ys))
print(cubica(xs, ys))

```

$y = 5.6867x + -0.8375$
 Error absoluto promedio: 17.833425742983504
 Desviación estándar: 13.065706140834441

Error máximo
 -5.0 -90.24

Error mínimo
 3.38 15.65

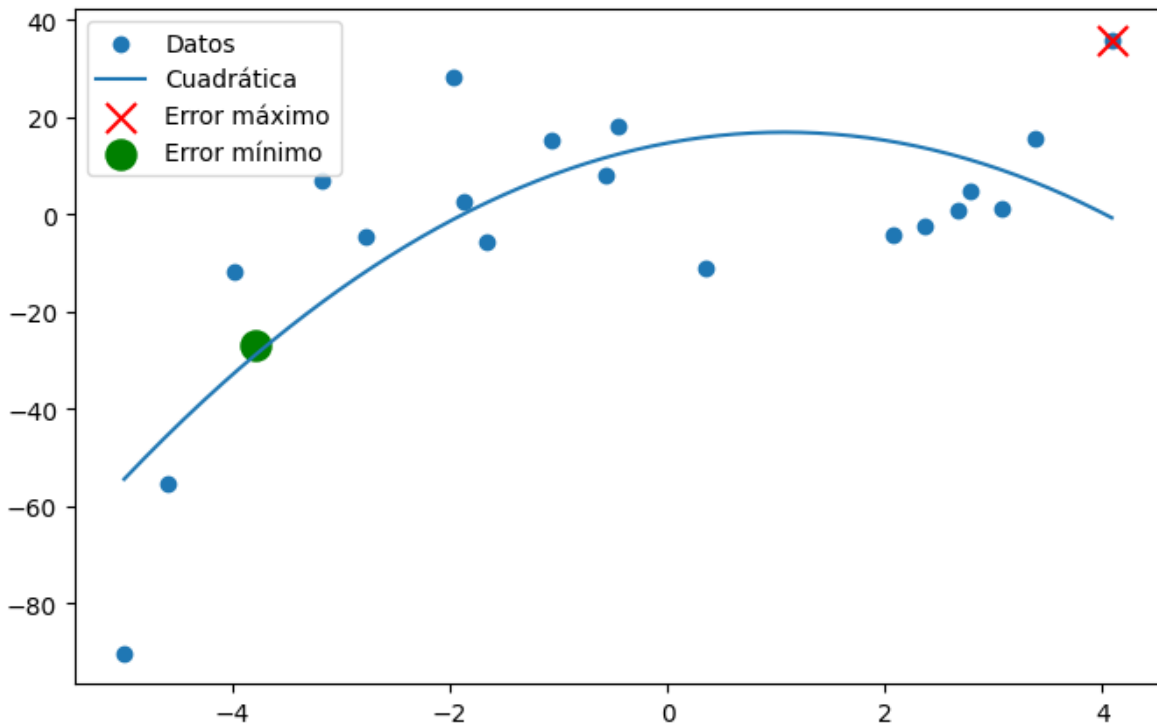


None

$y = -1.9347x^2 + 4.1525x + 14.7021$
 Error absoluto promedio: 14.53054767372313
 Desviación estándar: 10.737023909064197

Error máximo
 4.09 35.91

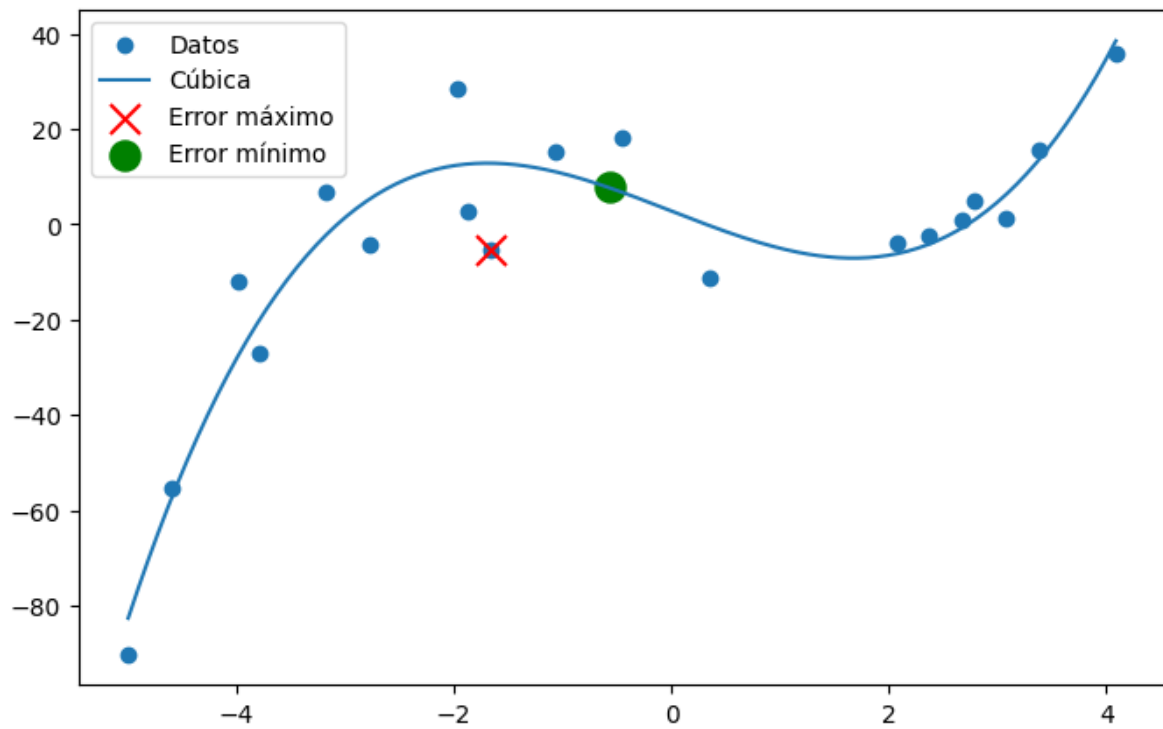
Error mínimo
 -3.79 -27.05



None
 $y = 1.0450x^3 + 0.0313x^2 + -8.8807x + 2.7623$
 Error absoluto promedio: 7.06197998516569
 Desviación estándar: 5.255180426741665

Error máximo
 -1.67 -5.47

Error mínimo
 -0.56 7.94



None